Experiment no-15

15. Implement probabilistic context-free grammar parsing for a sentence using python.

PROGRAM:

```python
import nltk
from nltk import PCFG
from nltk.parse import ViterbiParser

grammar = PCFG.fromstring("""
S -> NP VP [1.0]
NP -> 'ram' [0.5] | 'sita' [0.5]
VP -> V NP [1.0]
V -> 'likes' [1.0]
""")

parser = ViterbiParser(grammar)

sentence = input("Enter a sentence: ").lower().split()

try:
    for tree in parser.parse(sentence):
        print(tree)
except ValueError:
    print("Sentence contains words not in the grammar.")
```

OUTPUT:

```
Enter a sentence: ram likes sita
(S (NP ram) (VP (V likes) (NP sita))) (p=0.25)
```